

Online Examination Report

Date:	04.09.2014
Attempts:	1
Examinee:	Jackson, Bwayo
Date of birth:	1/20/1989
Subject:	080- Principles of flight
Result:	42,5 %
Time used:	00:30:00 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	230	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
2	217	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
3	232	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
4	184	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
5	221	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
6	214	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
7	203	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
8	149	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
9	209	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
10	212	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
11	199	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
12	192	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
13	147	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
14	185	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
15	243	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
16	146	81.	PRINCIPLES OF FLIGHT AEROPLANE	1,00	1,00
17	202	81.	PRINCIPLES OF FLIGHT AEROPLANE	0,00	1,00
18	113	81.1	SUBSONIC AERODYNAMICS	1,00	1,00
19	25	81.1.1.1	Laws and definitions	0,00	1,00
20	47	81.1.4.1	Streamline pattern	0,00	1,00
21	54	81.1.4.2	Induced drag	1,00	1,00
22	74	81.1.4.2	Induced drag	0,00	1,00
23	35	81.1.6	Ground effect	1,00	1,00
24	44	81.1.8.5	Special phenomena of stall	1,00	1,00
25	52	81.2.2.1	Normal shock waves	1,00	1,00
26	66	81.2.2.2	Oblique shock waves	0,00	1,00
27	59	81.2.2.2	Oblique shock waves	0,00	1,00
28	39	81.2.3.4	Effect on pitching moment	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	101	81.4	STABILITY	0,00	1,00
30	86	81.4.1.1	Basics and definitions	0,00	1,00
31	32	81.5.8.3	Stabiliser trim	0,00	1,00
32	6	81.6.1.1	Flutter	0,00	1,00
33	110	81.7	PROPELLERS	1,00	1,00
34	108	81.7	PROPELLERS	1,00	1,00
35	112	81.7	PROPELLERS	1,00	1,00
36	64	81.7.4.1	Torque reaction	1,00	1,00
37	81	81.7.4.4	Asymmetric blade effect	0,00	1,00
38	24	81.8.1.2	Straight steady climb	0,00	1,00
39	31	81.8.1.4	Straight steady glide	0,00	1,00
40	3	81.8.1.5	Steady coordinated turn	0,00	1,00
Total:42%				17,00	40,00

Attachment 2: List of result by topic

Licence

Date: 04.09.2014

Examinee: Jackson, Bwayo

Location: Uganda Civil Aviation Authority

Your Result: 17,00 from 40,00 Points (42,50 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
81. PRINCIPLES OF FLIGHT AEROPLANE	40	17,00	40,00	42,50
81.1. SUBSONIC AERODYNAMICS	7	4,00	7,00	57,14
81.2. HIGH SPEED AERODYNAMICS	4	1,00	4,00	25,00
81.4. STABILITY	2	0,00	2,00	0,00
81.5. CONTROL	1	0,00	1,00	0,00
81.6. LIMITATIONS	1	0,00	1,00	0,00
81.7. PROPELLERS	5	4,00	5,00	80,00
81.8. FLIGHT MECHANICS	3	0,00	3,00	0,00
Total	40	17,00	40,00	42,50

1

The wing of a weight shift aircraft twists so that the angle of attack

- ☒ from the center of the wing to the wing tip is variable and can be adjusted by the pilot in flight to optimize performance.
- ☐ changes from a high angle of attack at the center of the wing, to a low angle of attack at the tips.
- ☐ changes from a low angle of attack at the center of the wing, to a high angle of attack at the tips.

2

What is the relationship of false lift to the wind? False lift

- ☐ increases if the vertical velocity of a balloon increases.
- ☐ exists only if the surface winds are calm.
- ☒ decreases as the wind accelerates a balloon to the same speed as the wind.

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Subject: 080- Principles of flight
Date: 2014-09-04

3

During a 'no-gyro' approach and prior to being handed off to the final approach controller, the pilot should make all turns

☐ one-half standard rate unless otherwise advised.

☒ standard rate unless otherwise advised.

☐ any rate not exceeding a 30 degrees bank.

4

If the nosewheel of an airplane moves aft during gear retraction, how would this aft movement affect the CG location of that airplane? It would

- ☐ cause the CG location to move forward.
- ☒ have no effect on the CG location.
- ☐ cause the CG location to move aft.

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Date of birth: 1989-01-20

Subject: 080- Principles of flight
Date: 2014-09-04

5

Dampers should normally be kept closed during a climb to altitude because any air blown into the system would

- ☐ decrease the volume of gas within the envelope.
- ☐ increase the amount of gas to be valved, preventing the airship from ascending too fast.
- ☒ increase the amount of air to be valved, resulting in a slower rate of ascent.

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Date of birth: 1989-01-20

Subject: 080- Principles of flight
Date: 2014-09-04

6

What is the suggested speed to fly when passing through lift with no intention to work the lift?

- ☐ Best glide speed.
- ☒ Best lift/drag speed.
- ☐ Minimum sink speed.

7

When flying into a strong headwind on a long glide back to the airport, the recommended speed to use is the

- ☐ minimum sink speed.
- ☒ best lift/drag speed plus half the estimated windspeed at the glider's flight altitude.
- ☐ best glide speed.

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8

At what bank angle will the resultant of gravity and centrifugal force equal twice a glider's weight?

☐ 60 degree.

☒ 45 degrees.

☐ 30 degrees.

9

Unless adequate speed control is maintained during the turn to base and the final approach for a landing into the wind, which would most likely occur if a steep wind gradient existed?

- ☐ The desired landing spot would be undershot or the glider would stall.
- ☐ The wingtip on the outside of the turn would stall before the wingtip on the inside of the turn.
- ☒ The airspeed on final approach would increase, causing the glider to overshoot the desired landing spot.

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10

With regard to two or more gliders flying in the same thermal, which statement is true?

- ☐ Turns should be in the same direction as the highest glider.
- ☐ All turns should be to the right.
- ☒ Turns should be made in the same direction as the first glider to enter the thermal.

11

The best lift/drag ratio of a glider is a value that

- ☒ varies depending upon the weight being carried.
- ☐ remains constant and is independent of the weight being carried.
- ☐ remains constant regardless of airspeed changes.

12

An aircraft is loaded with the CG aft of the aft limit. What effect will this have on controllability?

- ☐ A stall will occur at a lower airspeed, but recovery will be easier because of reduced wing loading.
- ☒ Stall and spin recovery may be difficult or impossible.
- ☐ A stall will occur at a higher indicated airspeed due to the greater downloading on the elevator.

13

When a slight upward or negative flap deflection is used, the result is

☐ decreased drag.

☒ decreased lift.

☐ increased drag.

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14

If the landing gear on an airplane moves forward during retraction, the total moment will

☐ increase.

☐ remain the same.

☒ decrease.

15

(Refer to Figure 149.) What is the flight attitude? One system which transmits information to the instruments has malfunctioned.

See Attachments:

1. instrument_149.jpg

figure 149

(Attachment
No.1)

- ☐ Level turn to the left.
- ☒ Level turn to the right.
- ☐ Straight-and-level flight.

16

Which statement is generally true regarding wing camber of a glider's airfoil?

- ☐ The camber is the same on both the upper and lower surface of the wing.
- ☐ There is no camber on either the upper or lower surface of the wing.
- ☒ The camber is greater on the upper wing surface than it is on the lower surface of the wing.

17

What consideration should be given in the choice of a towplane for use in aerotows?

- ☐ Gross weight of the glider to be towed.
- ☐ Stall speed of the towplane.
- ☒ Towplane's low-wing loading and low-power loading.

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18

When the angle of attack of a symmetrical airfoil is increased, the center of pressure will

- ☐ remain unaffected.
- ☐ move aft along the airfoil surface.
- ☒ have very little movement.

19

Which of these statements about a stationary subsonic flow are correct or incorrect?

- I. The static pressure decreases as the streamlines converge.
II. The velocity increases as the streamlines converge.

- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☐ I is correct, II is incorrect.
- ☐ I is incorrect, II is incorrect.

20

Which of these statements about the strength of wing tip vortices are correct or incorrect?

- I. Assuming no flow separation, the strength of wing tip vortices decreases as the angle of attack decreases.
- II. The strength of wing tip vortices increases as the aspect ratio decreases.

- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

21

Winglets:

- ☐ create an elliptical lift distribution.
- ☒ decrease the induced drag.
- ☐ increase the manoeuvrability.
- ☐ decrease the static lateral stability.

22

An aeroplane transitions from steady straight and level flight into a horizontal co-ordinated turn with a load factor of 2, the speed remains constant and the:

- ☒ **total drag increases by a factor of 4.**
- ☐ **induced drag increases by a factor of 4.**
- ☐ **lift increases by a factor of 4.**
- ☐ **angle of attack increases by a factor of 1/4.**

23

Which statement, about an aeroplane leaving ground effect at constant angle of attack, is correct?

- I. The lift coefficient C_L remains constant.
II. The induced drag coefficient C_{Di} decreases.

- ☐ I is incorrect, II is correct.
- ☐ I is correct, II is correct.
- ☒ I is incorrect, II is incorrect.
- ☐ I is correct, II is incorrect.

24

Regarding deep stall characteristics, identify whether the following statements are correct or incorrect:

I. The combination of a wing with sweepback and a T-tail make an aeroplane prone to deep stall.

II. A stick pusher system can be fitted to an aeroplane that exhibits abnormal stall characteristics.

☐ I is incorrect, II is correct.

☐ I is incorrect, II is incorrect.

☐ I is correct, II is incorrect.

☒ I is correct, II is correct.

25

**Which of the following statements is correct?
Air passes a normal shock wave as**

☒ **The static temperature increases.**

☐ **The static temperature decreases.**

☐ **The velocity increases.**

☐ **The static pressure decreases.**

26

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The static temperature in front of an oblique shock wave is higher than behind it.
II. The static pressure in front of an oblique shock wave is lower than behind it.

- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

27

Which of these statements about an oblique shock wave are correct or incorrect?

- I. The static temperature behind an oblique shock wave is lower than in front of it.
II. The static pressure behind an oblique shock wave is higher than in front of it.

- ☐ I is incorrect, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is correct, II is correct.
- ☐ I is correct, II is incorrect.

28

Which of these statements about "tuck under" are correct or incorrect?

- I. A contributing factor to "tuck under" is a forward movement of the centre of pressure of the wing.
- II. A contributing factor to "tuck under" is an increase in the downwash angle at the location of the horizontal stabiliser.

- ☐ I is correct, II is correct.
- ☐ I is incorrect, II is incorrect.
- ☒ I is incorrect, II is correct.
- ☐ I is correct, II is incorrect.

29

If the aircraft's nose initially tends to move farther from its original position after the elevator control is pressed forward and released, the aircraft displays

☐ positive static stability.

☐ negative static stability.

☒ positive dynamic stability.

30

Positive static stability of an aeroplane means that following a disturbance from the equilibrium condition:

- ☐ the tendency is to move with an oscillatory motion of decreasing amplitude.
- ☐ the initial tendency is to diverge further from its equilibrium condition.
- ☒ the initial tendency is to return towards its equilibrium condition.
- ☒ the tendency is to move with an oscillatory motion of increasing amplitude.

31

When comparing a stabiliser trim system with an elevator trim system, which of these statements is correct?

- ☒ a stabiliser trim is able to compensate larger changes in pitching moments.
- ☐ an elevator trim is more suitable for aeroplanes with a large CG range.
- ☐ an elevator trim is able to compensate larger changes in pitching moments.
- ☒ a stabiliser trim is more sensitive to flutter.

32

Which of these statements about flutter are correct or incorrect?

I. If flutter occurs, IAS should be reduced.

II. Resistance to flutter increases with increasing wing stiffness.

☐ I is correct, II is correct.

☐ I is incorrect, II is incorrect.

☐ I is incorrect, II is correct.

☒ I is correct, II is incorrect.

33

The reason for variations in geometric pitch (twisting) along a propeller blade is that it

- ☐ prevents the portion of the blade near the hub to stall during cruising flight.
- ☐ permits a relatively constant angle of incidence along its length when in cruising flight.
- ☒ permits a relatively constant angle of attack along its length when in cruising flight.

34

Blade angle of a propeller is defined as the angle between the

- ☐ angle of attack and chord line.
- ☐ angle of attack and line of thrust.
- ☒ chord line and plane of rotation.

35

On a multiengine airplane, where the propellers rotate in the same direction, why is the loss of power on one engine more critical than the loss of power on the other engine?

- ☐ The torque reaction from operation of the critical engine is more severe around the vertical axis as well as the longitudinal axis.
- ☒ The asymmetric propeller thrust or P-factor results in the center of thrust from one engine being farther from the airplane centerline than the center of thrust from the other engine.
- ☐ The corkscrew pattern of airflow from one propeller is less effective against the airflow from the critical engine.

36

The torque reaction of a rotating fixed pitch propeller will be greatest at:

- ☐ low aeroplane speed and low engine power.
- ☒ low aeroplane speed and maximum engine power.
- ☐ high aeroplane speed and low engine power.
- ☐ high aeroplane speed and maximum engine power.

37

A propeller is turning to the right when viewed from behind; The asymmetric blade effect in the climb at low speed will:

- ☐ yaw the aeroplane to the right.
- ☒ roll the aeroplane to the left.
- ☐ yaw the aeroplane to the left.
- ☐ roll the aeroplane to the right.

38

An aeroplane climbs to cruising level with a constant pitch attitude and maximum climb thrust, (assume no supercharger).
How do the following variables change during the climb? (Gamma = flight path angle)

- ☐ Gamma decreases, angle of attack increases, IAS remains constant.
- ☒ Gamma remains constant, angle of attack remains constant, IAS decreases.
- ☐ Gamma decreases, angle of attack remains constant, IAS decreases.
- ☐ Gamma decreases, angle of attack increases, IAS decreases.

39

What factors determine the distance travelled over ground of an aeroplane in a glide from a given altitude?

- ☐ The wind and CLMAX.
- ☐ The wind and the aeroplane's mass.
- ☐ The wind and the lift/drag ratio.
- ☒ The wind and weight together with power loading, which is the ratio of power output to the weight.

40

An aeroplane performs a steady horizontal, co-ordinated turn with 45 degrees of bank at 230 kt TAS. The same aeroplane with the same bank angle and speed, but at a lower mass will turn with:

- ☒ a larger turn radius.
- ☐ a lower rate of turn.
- ☐ a smaller turn radius.
- ☐ the same rate of turn.

Attachment No. 1

Questions 15

figure 149

instrument_149.jpg

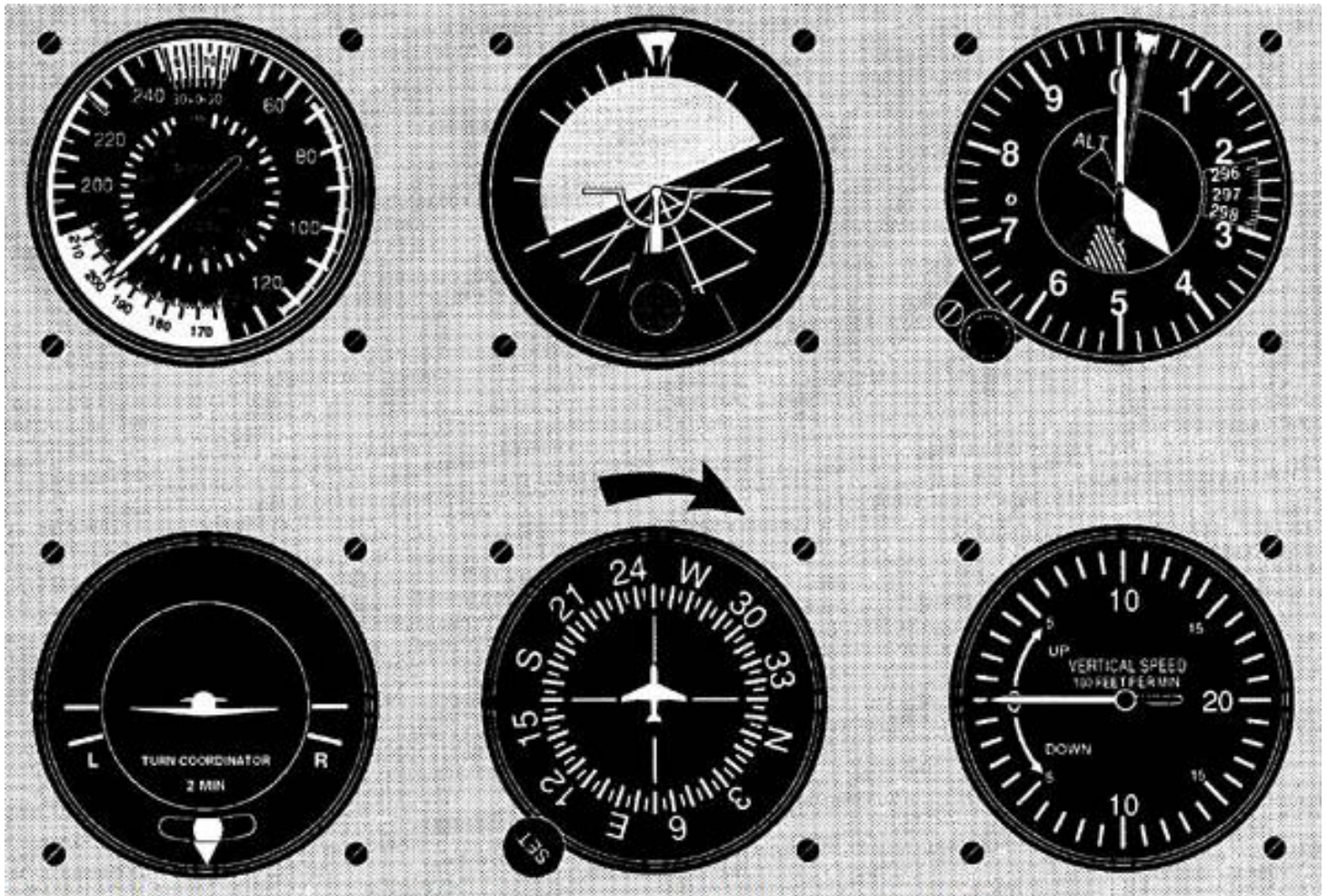


Figure 149. Instrument Interpretation (System Malfunction) © ASA